Status	Finished
Started	Saturday, 3 October 2026, 1:28 PM
Completed	Saturday, 3 October 2026, 1:38 PM
Duration	9 mins 54 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search continues in the right sublist
c. The search continues in the left sublist
d. The search ends successfully ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Linear search ✓
d. Jump search

Question **3**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It works on unsorted lists
d. It compares elements sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is the last element in the array

c. When the item is somewhere in the middle of the array ✓

d. When the item is not the array at all

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓

b. When the list is very small

c. When the list is unsorted

d. When the list contains duplicate elements

Question **6**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓

b. At the beginning of the list

c. In the left sublist

d. In the middle of the list

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element

b. Checking each element sequentially ✓

c. Dividing the list in half repeatedly

d. Sorting the list before searching

Question **8**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search can find multiple instances of the target element
- c. Binary search works on unsorted lists
- d. Binary search does not require dividing the list

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- d. Must use a sorted array

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 100
- b. 500
- c. 10 ✓
- d. 1000

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The higher middle element is chosen as the middle element
- b. The middle element is chosen randomly
- c. The lower middle element is chosen as the middle element ✗
- d. The search stops

Question **12**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Divide the list into two equal parts ✗
- b. Sort the list
- c. Compare the target element with the middle element in the list
- d. Compare the target element with the first element in the list

Question **13**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. n ✓
- c. n/2
- d. log n

Question **14**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Hash search
- c. Interpolation search
- d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the list contains duplicate elements
- c. When the element is found in the list ✓
- d. When the search algorithm finishes